



# Tackling the challenge facing PCB interconnect supplier

- Wendy Preston, Technical Marketing Engineer, Harwin

Too often media attention within the electronics business is focused almost totally on the IC technology involved and the issues that have to be dealt with in that respect. However, few recognise that exactly the same dynamics are at play elsewhere - in reality they have just as serious implications for connector manufacturers too, presenting difficult obstacles for them to overcome.

An increasing density of electronic systems is witnessed across a broad spectrum of applications. In some cases this is down to greater amounts of electronics content being implemented into such systems, for others it is due to the downsizing of designs, but frequently it is both of these at once. From instance, systems developed for use in UAVs, as well as in an automotive and space context need to fit all the supporting electronics into highly compact enclosures, that leaves little room for the connectivity aspect to be covered. The rising number of ICs utilised (and the heatsinking that must accompany them) means there is very little area available on the PCB. Furthermore, those PCBs will be packed closer together. That said, though OEMs want PCB connectors with low profiles (and reduced overall dimensions in general) from their suppliers, they cannot accept any sort of downgrading in the operational performance. The connectors still need to be able to cope with elevated current and exhibit tighter tolerances.

Industrial automation is a good example of some of the pressures involved. With the advent of Industry 4.0, the level of complexity going into systems developed for such purposes is

ramping up rapidly. Nevertheless there is still a need to adhere to space constraints, so that they don't hinder the smooth running of the facility into which they have been deployed. Connectors must occupy minimal space, but at the same time there is an expectation that there won't be any compromises with regard to reliability, as the costs associated with downtime are too great. If a factory production line is halted for several hours due to the failure of an electrical connection it could result in many thousands of Euros being lost.

Another concern for connector suppliers is that when it comes to designing the connectivity element into a system, engineering teams don't always think far enough ahead. Too often this part of the development project isn't judged to be a priority. It is left to the last minute, and then there is insufficient space left available. This puts huge pressure on the connector supplier to offer more compact solutions that still adhere to the same performance characteristics. On top of this OEMs still want to keep the total bill-of-materials costs low and also make assembly as simple as possible.

It may seem obvious, but by listening to our customers and getting insight into what they need, we are able to continually bring out innovative products and consequently grow our market share. For example, Harwin's connectors are used by many of the leading motorsport teams around the world. The demanding nature of this sort of application relies on constant dialogue between our engineering team and the customer - as they look to reduce weight, save space and keep on pushing performance envelope. This is a

philosophy that we maintain throughout dealings with our entire customer base and it is allowing us to keep up with ever more ambitious industry demands. In addition, we are also regularly upgrading our manufacturing facilities to make sure that we have the cutting-edge equipment needed to produce next generation connectivity solutions. Harwin also works closely with a number of universities in the UK, continental Europe and North America on emerging applications that will impact on the industry in the longer term. The combination of these factors is allowing us to get the lead on our competition.

Harwin's Gecko family of 1.25mm pitch connectors is really where we have put all the different facets of our business that were just discussed into practice. These units meet the industry's need for higher density PCB interconnects (with each contact able to support 2A currents), while being both lighter and more cost effective than rival solutions (thanks to their plastic housings). Their robust construction means that they can cope with high degrees of shock and vibration, as well as extreme temperatures. Screw based connection options make them more reliable in challenging applications environments compared to conventional latch-based solutions. Since the release of the first products in this family in late 2012, the market uptake has been impressive and is already representing a large proportion of our high reliability connector shipments. As well as all the other applications already mentioned, these products are proving highly popular in the defence and robotics sectors.

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